

Secrets of Handheld Interface Design

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Handspring, Inc.



Agenda

- Handheld UI requirements
- Wireless UI requirements
- Understanding our customers
- Three generations of UI philosophies
- Five questions I'll ask about your UI



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Handheld UI requirements

“Mobile” interface design

Common misconceptions about UI

- Window dressing
- More cosmetic than functional
- Implemented at the end of the project



Handspring's approach

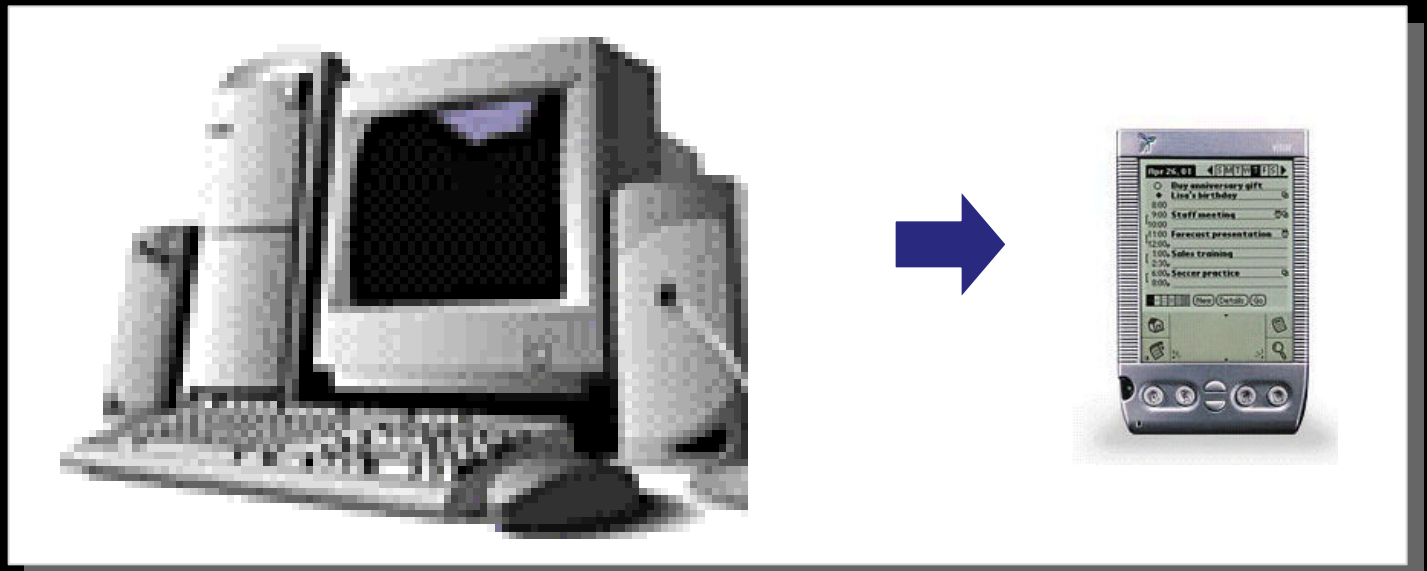
- Core to the design of the application
- Choose the right feature set to do a task
- Implemented at the beginning of a project

Arranging pixels to *solve problems*



Handheld challenges

- Limited screen space & power
- Mobility = urgency
- How do you design for smaller products?



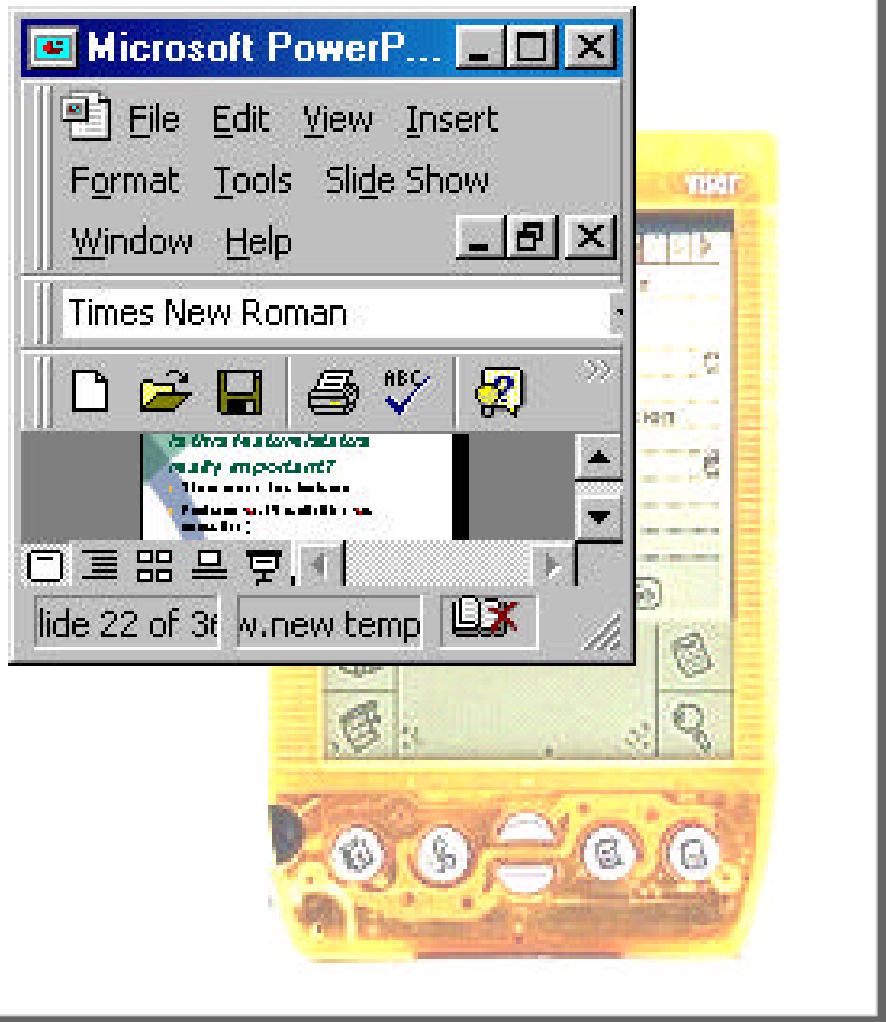
Hardware miniaturization

- Sony® Walkman®
- Tape recorder that doesn't record?
- No speakers
- Clear target: personal, portable music
- Unorthodox optimization



Software miniaturization

- Disciplined tradeoffs
- Strict optimization
- Focus on solving problems, not adding features



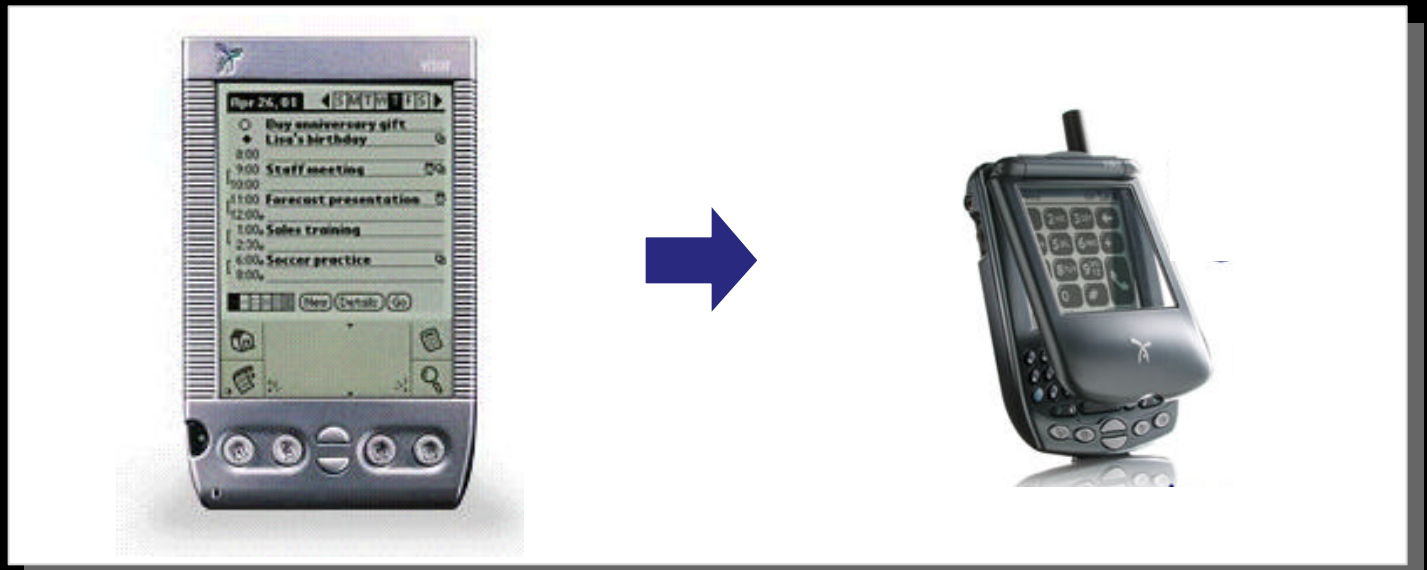


Wireless UI requirements

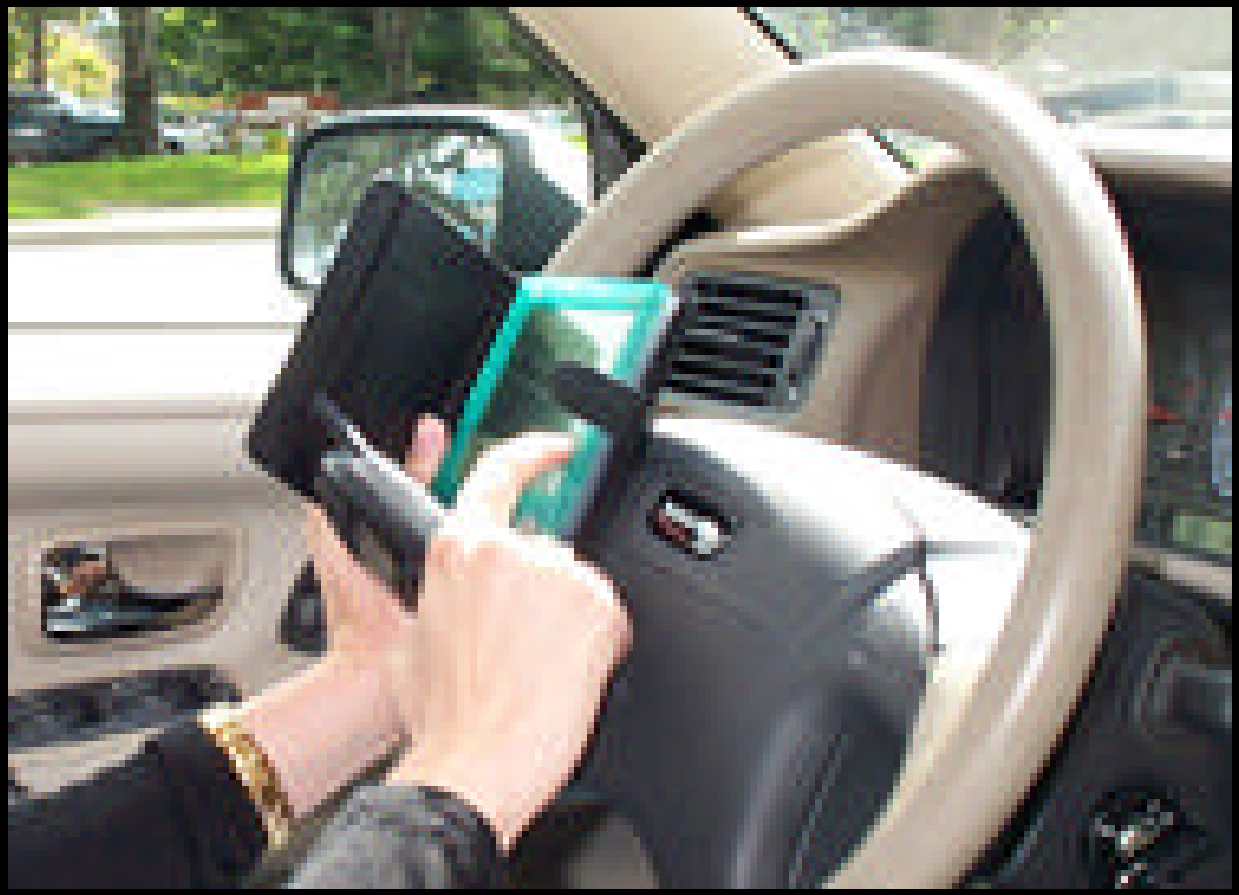
New game, new rules—again

Wireless UI challenges

- Greater urgency (e.g., answering the phone)
- Tighter screen space and power constraints
- Different networks and requirements
- Tight integration of functionality



Be afraid



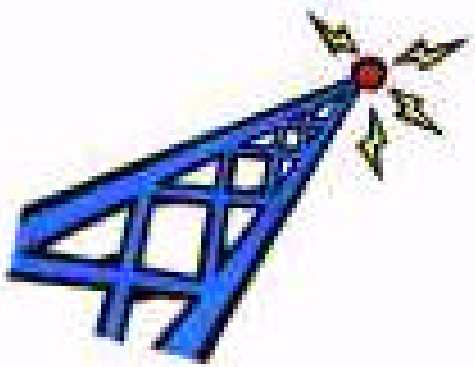
Lessons learned: one-handed operation

- Keyboard is better than stylus
- Big buttons are better than small buttons
- People don’t like to tap on the screen



Lessons learned: network requirements

- Dial-in issues
  - Pain of establishing connection
  - Power
  - Cost
  - Dropping the session
  - Can’t assume packet-only
- Latency and throughput
  - Asynchronous vs. synchronous design
  - Synchronous vs. pseudo-synchronous design
  - Status and expectations





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Understanding our customers

Who’s problems are we trying to solve?

Who we are

- We are geeks
- The tools we use = our job expertise
- This is a great coincidence (for us…)!



=



Our job focus

Our tools

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Who our customers are

- Customers forced to use computers regardless of expertise
- Result: anxiety, stress (conduct user tests!)
- Market growth driven by “non-geek” consumers



Customer’s job focus

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Customer’s tools

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Imagine trading places...

- Think of something you hate using
- What if you *must* use those tools for your job
- How complex a tool would you want?



Our job focus
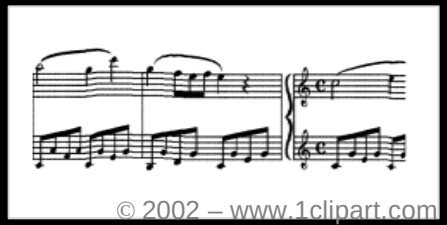
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Our tools

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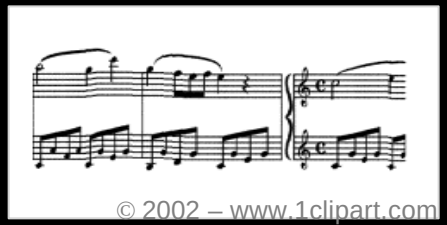
Opposite attitudes





	Us (Geeks)	Our customers
Technology	**Use whenever possible**	**Avoid unless necessary**
Status	Exactly what's going on?	Don't care unless there's a problem
Functional objectives	Do more, efficiently	Just get the job done
Rich feature set =	More control	Unpredictable (less control!)
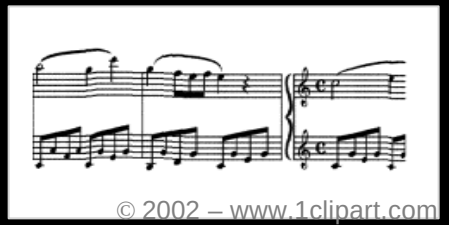
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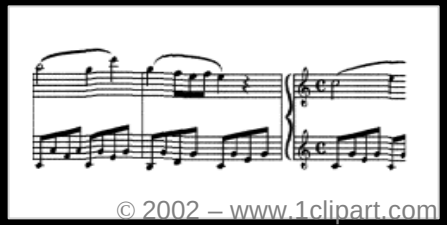
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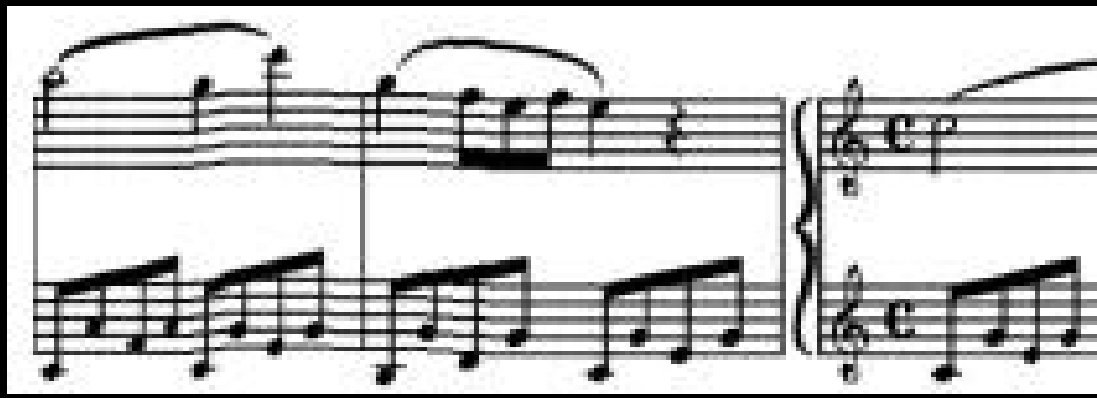
Simplicity wins

- Most people don’t use most features
- Robust functionality can cripple usability
- “Pack light”



What about power users?

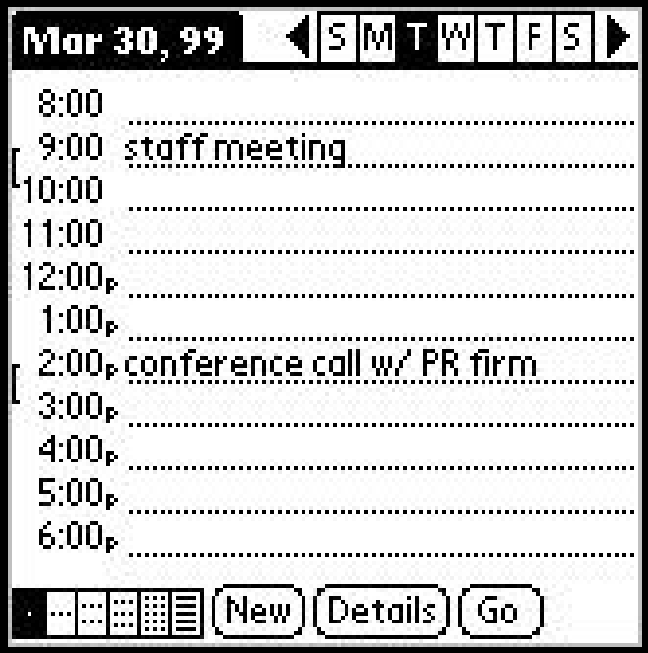
- Power users like ease of use too
- Sometimes you have to appeal to power users
- “Provide but hide” (from novice customers)



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Example: Datebook Plus

- Power features
  - Floating appointments
  - Integrated To Do items
  - Five views
  - Templates
  - Journal, etc.
- Most of it is “hidden”
- You need to add features as markets mature





Three generations of UI philosophies

"The Road goes ever on and on"

UI design philosophy, 1996

- “Connected organizer”
- New market, limited competition
- The need to simplify
- *Less is more*



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UI design philosophy, 2000

- “Handheld computer”
- Increased competitive pressures
- The need to innovate and evolve
- *Simplicity is a means, not an end*



UI design philosophy, 2002

- “Communicator”
- New markets, but competition rushing in
- The need to simplify converging technologies
- *Simplify vs. Simple*





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Five questions I'll ask about your UI

Doing the right thing

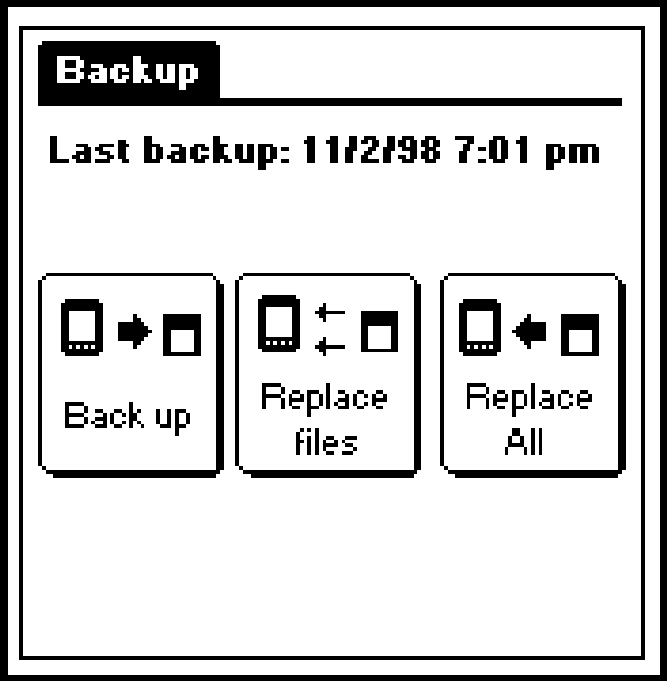
1. What is your customer trying to do?

- Optimize for frequent actions
  - Put high priority features up front
  - Hide or remove everything else
- Design decisions depend on usage patterns

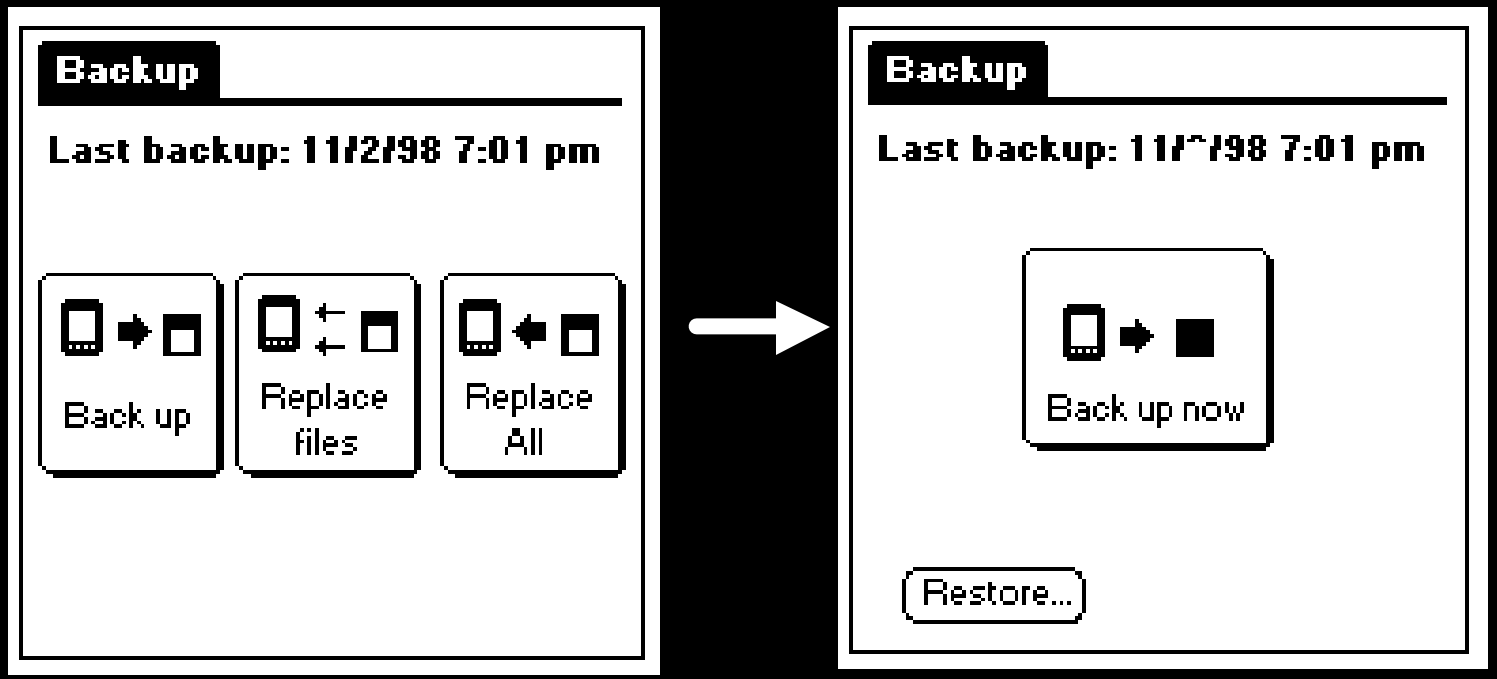


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Optimization: example

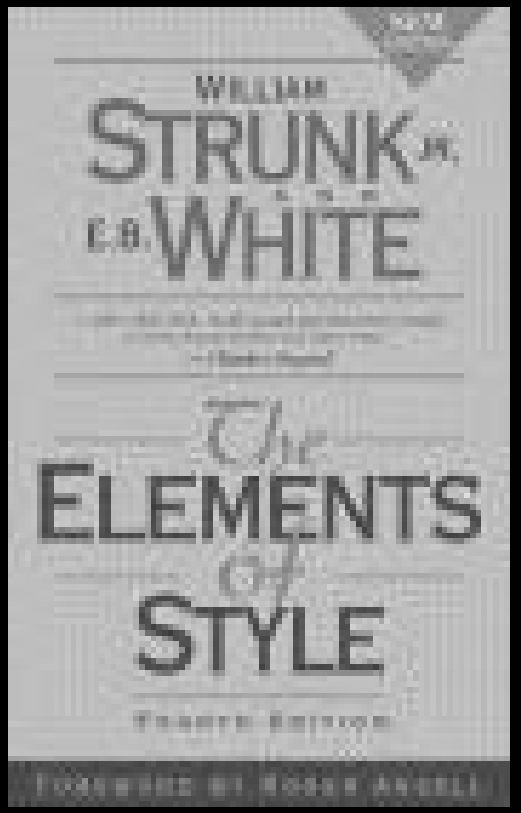


Optimization: example



2. Is this feature really important?

- “There are no bad features”
- Innovative products don’t need bells and whistles
- “Strunk & White approach”



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Strunk & White Example

> “ Style guides must go beyond the mere definition of interface components and usage but must abstract to higher levels and include emerging patterns of interaction.”

David Anderson, uidesign.net



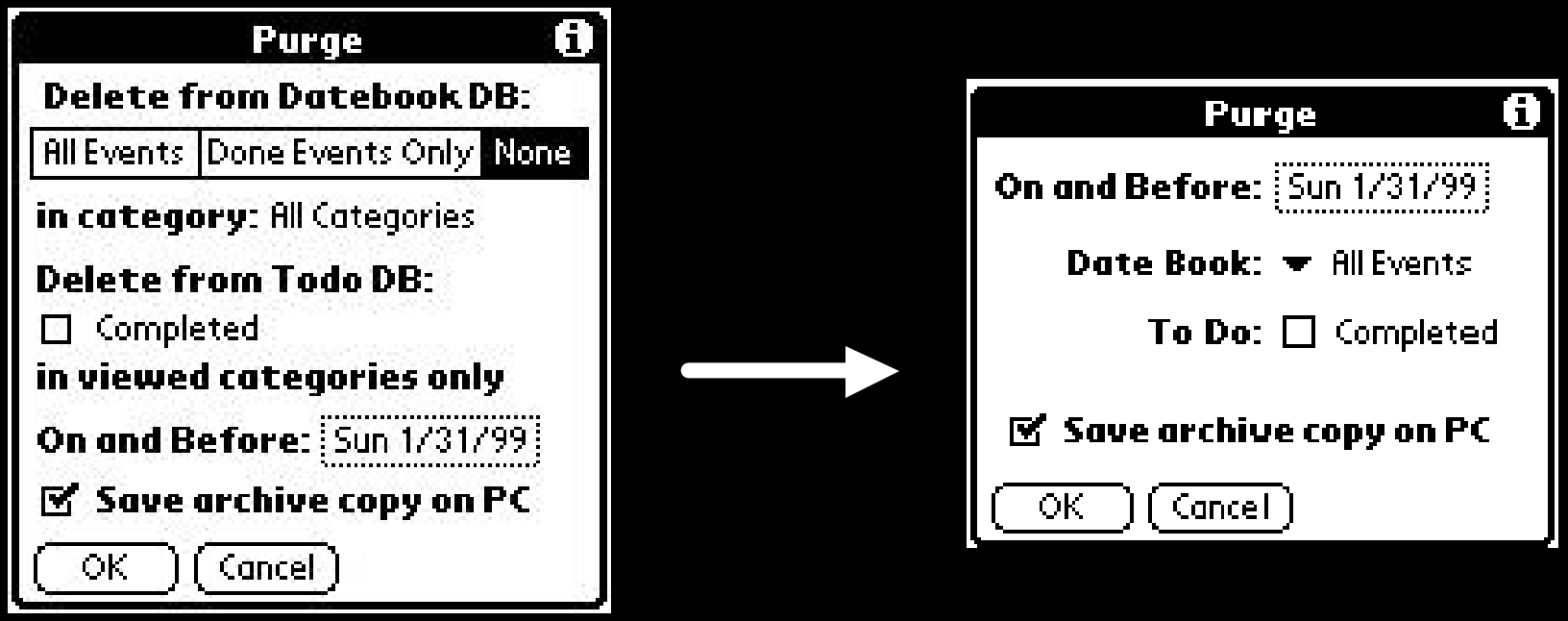
Strunk & White Example

“Do the right thing.”

Applying Strunk & White

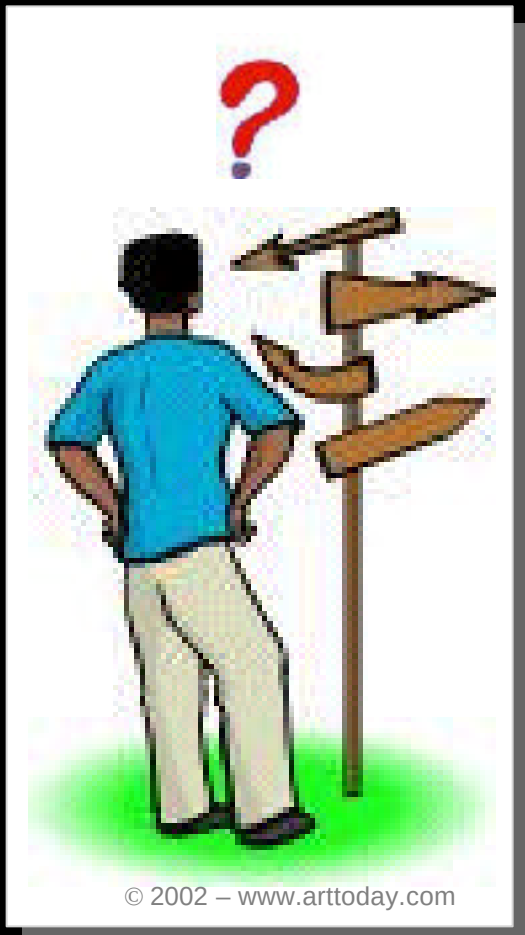


Applying Strunk & White



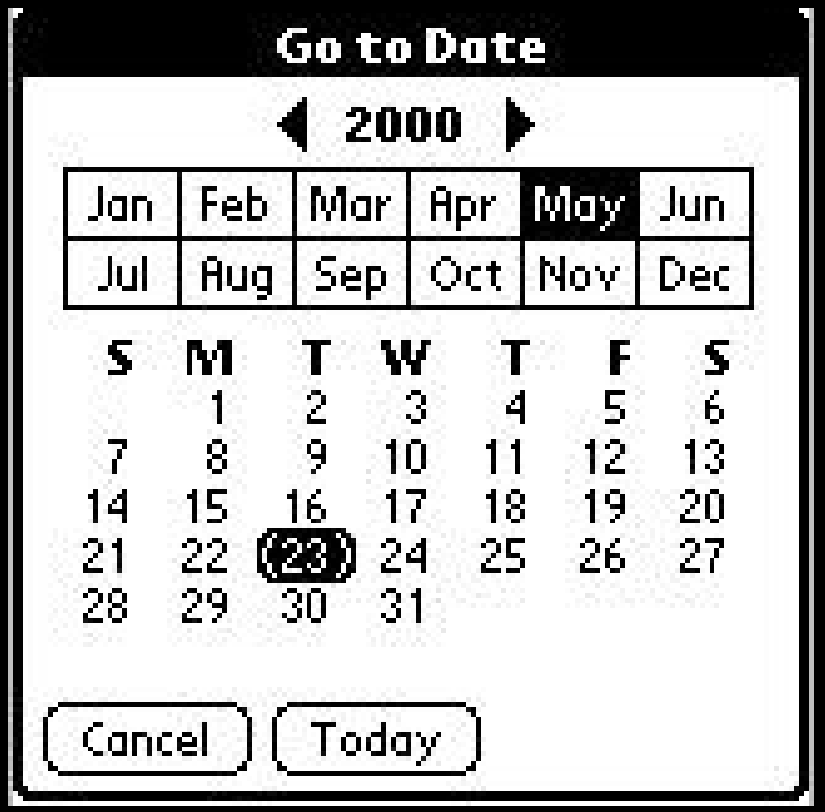
3. Why not use a standard UI object?

- Intuitive = don’t have to think
- “UI déjà vu”
- Predictable > cool



Examples

- Icons vs. buttons
- Graffiti indicator placement
- Tab interfaces
- *OK to break rules if you have a reason*



4. Do you have time to put all this in?

- Features are less important than we think
- Features take more time than we think
- When to sacrifice good features
  - Be realistic up front
  - Be ruthless at the end ("Kowalski in the boiler room")



5. Do you want to keyboard-enable your app?

- One-handed access to priority features
- Keyboard access to other features
- Minimize need for stylus but must provide stylus access



Keyboard tips



- Provide focus for on-screen buttons
- Use Space bar as an “enter” key
- Scroll behavior (line vs. page)
- Use backspace and return
- Add menu commands for command buttons
- Map prev/next field command to scroll buttons
- Don’t forget the app button
- Note: don’t need to get pathological

Conclusion: evaluating UI

- Did you delight the customer?
- “User” vs. “Customer”
  - User = no implied relation to designer
  - Customer = someone you want to please
- Delight = make someone smile
- None of the “rules” matter if you accomplish this



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For more information

- *Information Appliances and Beyond*, Edited by Eric Bergman
- *Designing Visual Interfaces*, Kevin Mullet and Darrell Sano
- *Piloting Palm* Andrea Butter and David Pogue



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